Learn C++ Programming -Beginner to Advance- Deep Dive in C++

Section 5:C++ Basics

Lecture 26. Variables

1. Introduction

Variables are the names given to data.

A variable has a data type.

A variable is a place where data is stored in a program.

If we have data and we want to process that data, we need to keep the data somewhere. For this, variables are used.

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2. Variable Concept

In mathematics also, variables are used.

Example:

$F(x) = 3x^2 + 4x$

Here, x is a variable.

Whatever value x has, we substitute that value and get the answer.

The same concept is used in programming also.

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3. Variable Must Have Data Type

A variable should have a data type.

Data type tells what type of value the variable can store.

Example:

If we want to store the roll number of a student, the roll number is integer type.

A roll number will not have a decimal point.

Example:

Roll number = 2.5

This is not correct because roll number is an integer value.

Roll number will also not be negative.

Roll numbers are usually one-digit, two-digit, or three-digit numbers.

So, integer type can be used.

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4. Declaring a Variable

Example:

```
int rollNumber;
```

Here:

int

is the data type.

rollNumber

is the variable name.

This is the method of declaring a variable.

A variable must be declared before using it in the program.

Every variable must be declared first, and then only it can be used.

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5. Memory for Variable

When a variable is declared and the program runs, memory is occupied at runtime.

Example:

```
int rollNumber;
```

If integer takes 2 bytes, then `rollNumber` will consume 2 bytes of memory.

If integer takes 4 bytes, then it will consume 4 bytes of memory.

Every byte in memory has an address.

Example:

```
200
201
```

These can be memory addresses of the bytes used by the variable.

Any number stored in the variable is stored in memory at runtime.

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6. Assigning Value to Variable

After declaring a variable, a value can be stored in it.

Example:

```
rollNumber = 10;
```

Here, the value `10` is stored in the variable `rollNumber`.

Before assigning `10`, the variable may contain a garbage value.

Garbage value means an unknown value.

It is already present, but we have not stored it.

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7. Declaration and Initialization

Example:

```
int rollNumber;
```

This is called declaration of a variable.

Example:

```
rollNumber = 10;
```

This is called initialization of a variable.

When a variable is declared and the program runs, it occupies memory, and the value is stored in it.

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8. Declaration with Initialization

A variable can be declared and initialized at the same time.

Example:

```
int rollNumber = 10;
```

Here, the variable is declared as well as initialized.

If a variable is initialized at the time of declaration, it does not get a garbage value.

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9. Character Variable

If we want to store a section or group like Group A, Group B, or Group C, then we can use character type.

Example:

```
char group = 'A';
```

Character values must be enclosed in single quotes.

Example:

```
'A'
```

Every character type value should be enclosed in single quotes.

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10. Float Variable

If we want to store the price of an item, such as:

12 dollars 75 cents

then we can use float type.

Example:

```
float price = 12.75f;
```

It is better to write f after the decimal value to show that it is a float value.

If f is not written, the value is assumed as double by default.

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11. Examples of Variable Declaration and Initialization

```
int rollNumber = 10;
char group = 'A';
float price = 12.75f;
```

These examples show integer type, character type, and float type variable initialization.

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12. Valid and Invalid Variable Names

Some variable names are valid and some are invalid.

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13. Number at the Beginning Is Not Allowed

Valid:

```
int x1;
```

Invalid:

```
int 1x;
```

A number at the beginning of a variable name is not allowed.

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14. Spaces Are Not Allowed

Valid:

```
int rollno;
```

Invalid:

```
int roll number;
```

There should not be any space in a variable name.

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15. Using Underscore

If we want to read a variable name as two words, we can use underscore.

Example:

```
int roll_number;
```

This is allowed.

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16. Using Capital Letters in Variable Names

Another way of writing variable names is:

```
int rollNo;
```

Here, `roll` is written in small letters and `No` starts with a capital letter.

Another form:

```
int RollNo;
```

This notation can also be followed.

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17. Meaningful Variable Names

Variable names should be meaningful.

Avoid names like:

x1
x2
abc

Use meaningful names such as:

price
group
rollNumber

Meaningful names are readable and show what type of value the variable is containing.

This makes writing the program easier.

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18. Key Points

Variables are names given to data.

A variable has a data type.

A variable is used for storing data.

A variable must be declared before using it.

At runtime, a variable occupies memory.

A variable may contain garbage value before initialization.

Initialization means storing a value in the variable.

A variable can be declared and initialized at the same time.

Character values are written in single quotes.

Float values can use f.

A variable name cannot start with a number.

Spaces are not allowed in variable names.

Underscore can be used in variable names.

Meaningful variable names should be used.